

# Algebra I — Solving Quadratics by Factoring

Teacher Packet • Monday (Launch Day) • everything on one page to review

## Part 1 Slides (project these)

### Slide 1 / 9 • Hook Why solve for zero?

- A ball's height is  $h = -16t^2 + 64t$ . When does it hit the ground?
- That's a **quadratic** — and we need to find WHEN  $h = 0$ .
- Graphing works but is slow — there's a faster algebra trick this week.
- By Friday you'll solve this in under 30 seconds.

### Slide 2 / 9 • Big picture This week

- A quadratic can have 0, 1, or 2 solutions (roots / zeros).
- Factoring turns one hard equation into two easy ones.
- **Key idea:** if two things multiply to 0, one of them must BE 0.
- Mon  $a=1$  Tue  $a \neq 1$  Wed/Thu special cases + mixed Fri quiz.

### Slide 3 / 9 • Vocabulary Today's terms

- **Quadratic equation:** form  $ax^2 + bx + c = 0$ .
- **Root / zero / solution:** an  $x$  that makes it true.
- **Zero Product Property:** if  $A \cdot B = 0$  then  $A = 0$  or  $B = 0$ .
- **Factor:** rewrite as a product of simpler pieces.

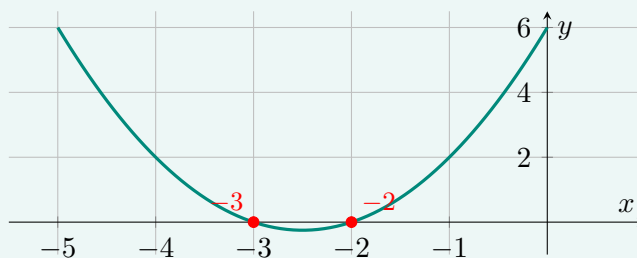
### Slide 4 / 9 • I do Model problem — fully worked

Solve  $x^2 + 5x + 6 = 0$ .

- Two numbers multiply to 6, add to 5  $\Rightarrow$  2 and 3.
- Factor:  $(x + 2)(x + 3) = 0$ .
- Zero Product Property:  $x + 2 = 0$  or  $x + 3 = 0$ .
- **Solutions:**  $x = -2$  or  $x = -3$  (check by substituting).

Slide 5 / 9 • See it

The roots are where the parabola crosses the  $x$ -axis



$y = x^2 + 5x + 6$  crosses at  $x = -3$  and  $x = -2$  — the same answers we factored.

Slide 6 / 9 • We do #1

Together — whiteboards up

Solve  $x^2 + 7x + 10 = 0$ .

$$(x + 2)(x + 5) = 0 \Rightarrow x = -2, -5$$

Slide 7 / 9 • We do #2

Different signs — think carefully

Solve  $x^2 - 2x - 15 = 0$ .

$$(x - 5)(x + 3) = 0 \Rightarrow x = 5, -3$$

Slide 8 / 9 • We do #3

No middle term — special case

Solve  $x^2 - 9 = 0$  (difference of squares).

$$(x - 3)(x + 3) = 0 \Rightarrow x = 3, -3$$

Slide 9 / 9 • Roadmap

Where we're going

- **Today:** factor when  $a = 1$ , Zero Product Property.
- **Tue:** factor when  $a \neq 1$  (e.g.  $2x^2 + 7x + 3$ ).
- **Wed/Thu:** special cases + mixed practice.    **Fri:** quiz.

## Part 2 Group examples — worked for you

1.  $x^2 + 7x + 10 = 0$ : product 10, sum 7  $\rightarrow 2, 5$ ;  $(x + 2)(x + 5) = 0$ ;  $x = -2, -5$ .
2.  $x^2 - 2x - 15 = 0$ : product  $-15$ , sum  $-2 \rightarrow -5, 3$ ;  $(x - 5)(x + 3) = 0$ ;  $x = 5, -3$ .
3.  $x^2 - 9 = 0$ : difference of squares;  $(x - 3)(x + 3) = 0$ ;  $x = 3, -3$ .

## Part 3 Guided notes (teacher key — blue = answers)

### Vocabulary

**Zero Product Property:** if  $A \cdot B = 0$ , then  $A = 0$  or  $B = 0$ .

**Root (solution):** an  $x$  that makes the equation true; on a graph it is an  $x$ -intercept.

**We do together.** Solve  $x^2 + 5x + 6 = 0$ .

**Step 1.** One side already = 0. Good.

**Step 2.** Factor:  $x^2 + 5x + 6 = (x+2)(x+3)$ .

**Step 3.** Set each factor = 0:  $x+2 = 0$  or  $x+3 = 0$ .

**Step 4.** Solve:  $x = -2$  or  $x = -3$ .

**You Try:** a)  $x^2 - x - 12 = 0$   $(x-4)(x+3) = 0 \Rightarrow x = 4, -3$ ; b)  $x^2 - 9 = 0$   $(x-3)(x+3) = 0 \Rightarrow x = 3, -3$ .

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## Part 4 Worksheet (student copy)

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**Final answers** (show your work):

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_ 4. \_\_\_\_\_ 5. \_\_\_\_\_ 6. \_\_\_\_\_ 7. \_\_\_\_\_ 8. \_\_\_\_\_ 9. \_\_\_\_\_ 10. \_\_\_\_\_  
11. \_\_\_\_\_

**Di-**

**Directions:** Solve each by factoring. State both roots.

1.  $x^2 + 5x + 6 = 0$

6.  $x^2 - 6x + 8 = 0$

2.  $x^2 + 7x + 10 = 0$

7.  $x^2 + x - 12 = 0$

3.  $x^2 - 2x - 15 = 0$

8.  $x^2 - 25 = 0$

4.  $x^2 - 9 = 0$

9.  $x^2 - 4x = 0$

5.  $x^2 + 8x + 16 = 0$

10.  $x^2 + 2x - 8 = 0$

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**Challenge** (preview of  $a \neq 1$ ):

11.  $2x^2 + 5x - 3 = 0$

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## Part 5 Answer key — fully worked

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**Procedure:** make one side = 0, factor, set each factor to 0, solve.

1.  $(x+2)(x+3) = 0 \Rightarrow x = -2, -3$

2.  $(x+2)(x+5) = 0 \Rightarrow x = -2, -5$

3.  $(x-5)(x+3) = 0 \Rightarrow x = 5, -3$

4.  $(x-3)(x+3) = 0 \Rightarrow x = 3, -3$

- 5.  $(x + 4)^2 = 0 \Rightarrow x = -4$  (double root)
- 6.  $(x - 2)(x - 4) = 0 \Rightarrow x = 2, 4$
- 7.  $(x + 4)(x - 3) = 0 \Rightarrow x = -4, 3$
- 8.  $(x - 5)(x + 5) = 0 \Rightarrow x = 5, -5$
- 9.  $x(x - 4) = 0 \Rightarrow x = 0, 4$
- 10.  $(x + 4)(x - 2) = 0 \Rightarrow x = -4, 2$
- 11.  $2x^2 + 5x - 3 = (2x - 1)(x + 3) = 0 \Rightarrow x = \frac{1}{2}, -3$